

Day 19

Even and Odd Parity Generation in Verilog

What is a Parity Bit?

A parity bit is an extra bit added to a group of binary data to make the total number of 1s either even or odd.

- Even Parity Bit: Ensures the total number of 1s (including the parity bit) is even.
- Odd Parity Bit: Ensures the total number of 1s (including the parity bit) is odd.

Verilog Code

A) Even parity generator

```
1 module even_parity_gen(in,p );
2 input [3:0]in;
3 output p;
4 assign p=in[3]^in[2]^in[1]^in[0];
5 endmodule
```

B) Odd parity generator

```
1 module odd_parity_gen(in,p);
2 input [3:0]in;
3 output p;
4 assign p=~(in[3]^in[2]^in[1]^in[0]);
5 endmodule
```

Testbench Code

A) Even parity generator tb

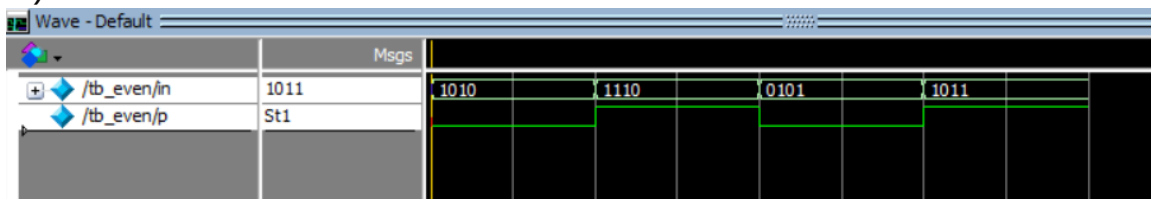
```
1 module tb_even;
2   reg[3:0] in;
3   wire p;
4   even_parity_gen dut(.p(p),.in(in));
5   initial
6   begin
7     in=4'b1010; #10
8     in=4'b1110; #10
9     in=4'b0101; #10
10    in=4'b1011; #10
11    $finish;
12  end
13 endmodule
14
```

B) Odd parity generator tb

```
1 module tb_odd();
2   reg[3:0] in;
3   wire p;
4   odd_parity_gen dut(.p(p),.in(in));
5   initial
6   begin
7     in=4'b1010; #10
8     in=4'b1110; #10
9     in=4'b0101; #10
10    in=4'b1011; #10
11    $finish;
12  end
13 endmodule
14
```

Waveform

A)

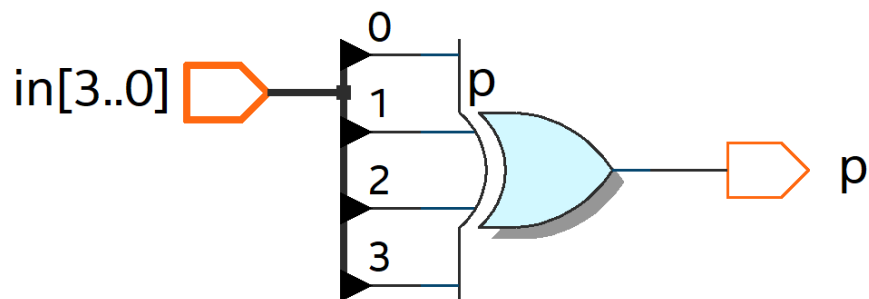


B)

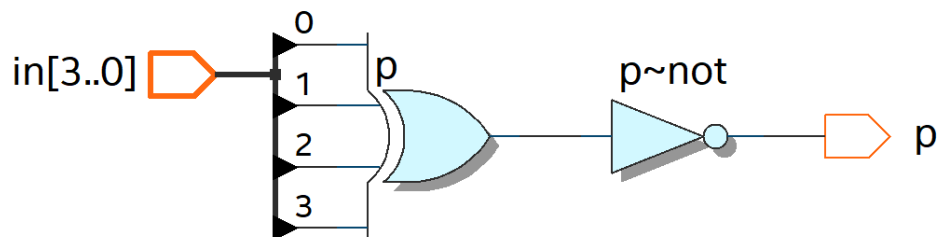
Wave - Default		Msgs									
+	/tb_odd/in	1011	1010		1110		0101		1011		
	/tb_odd/p	St0									

Schematic

A)



B)



EDA Tools Used

- IntelQuartusPrime
- ModelSim